Set Operations

A set is a container for a collection of unique or distinct **elements**, without order.

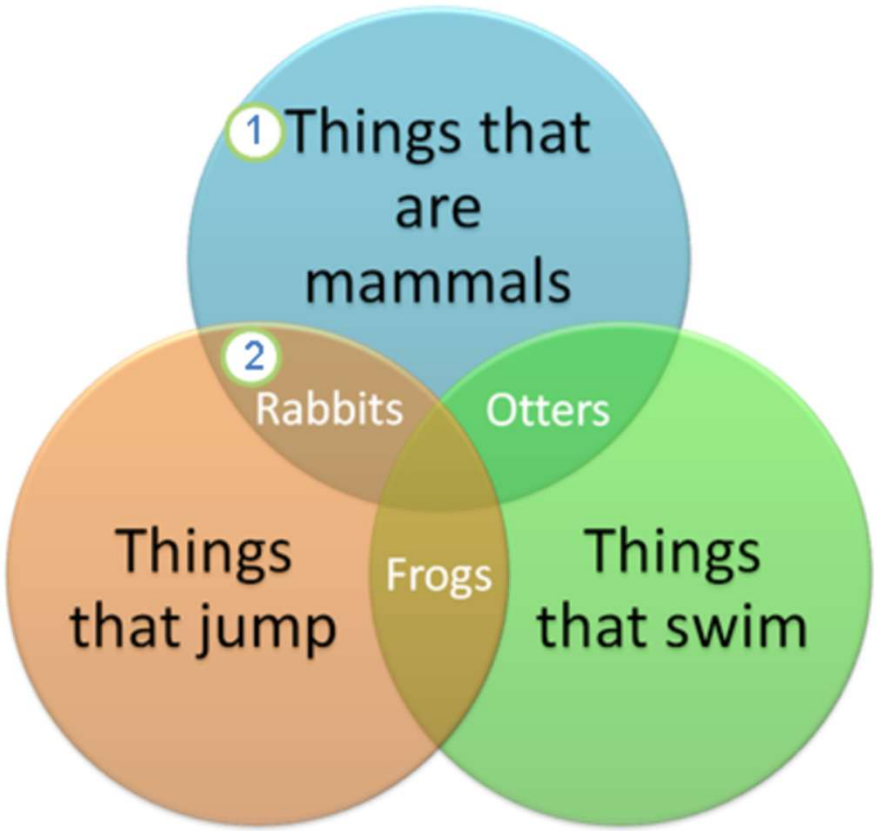
Order doesn't matter in a set; it is all about being a **member** or not.

Elements of a set do not have to be of the same type from a data science perspective. {cat, 1, blue}

We can do set operations on sets.

We can use symbols and Venn Diagrams for set operations.

Venn Diagram uses circles to represent sets. You can make your own in PowerPoint, but I copied this from Create a Venn diagram - Office Support (microsoft.com)



Dolphins and Orcas could be at the intersection of all three sets, as they swim, are mammals, and can jump out of the water. Can you think of others?

Review:

Nominal data is names. We can categorize, collect, describe, count, and do set theory.

Set Theory: A set is a container for a group of unique members or elements. Unique means that there are no duplicates or repeats. Order doesn't matter for a set.

Curly brackets are used in math to denote a set and commas are used to separate the elements. { Jae, Doc, Aury} { }

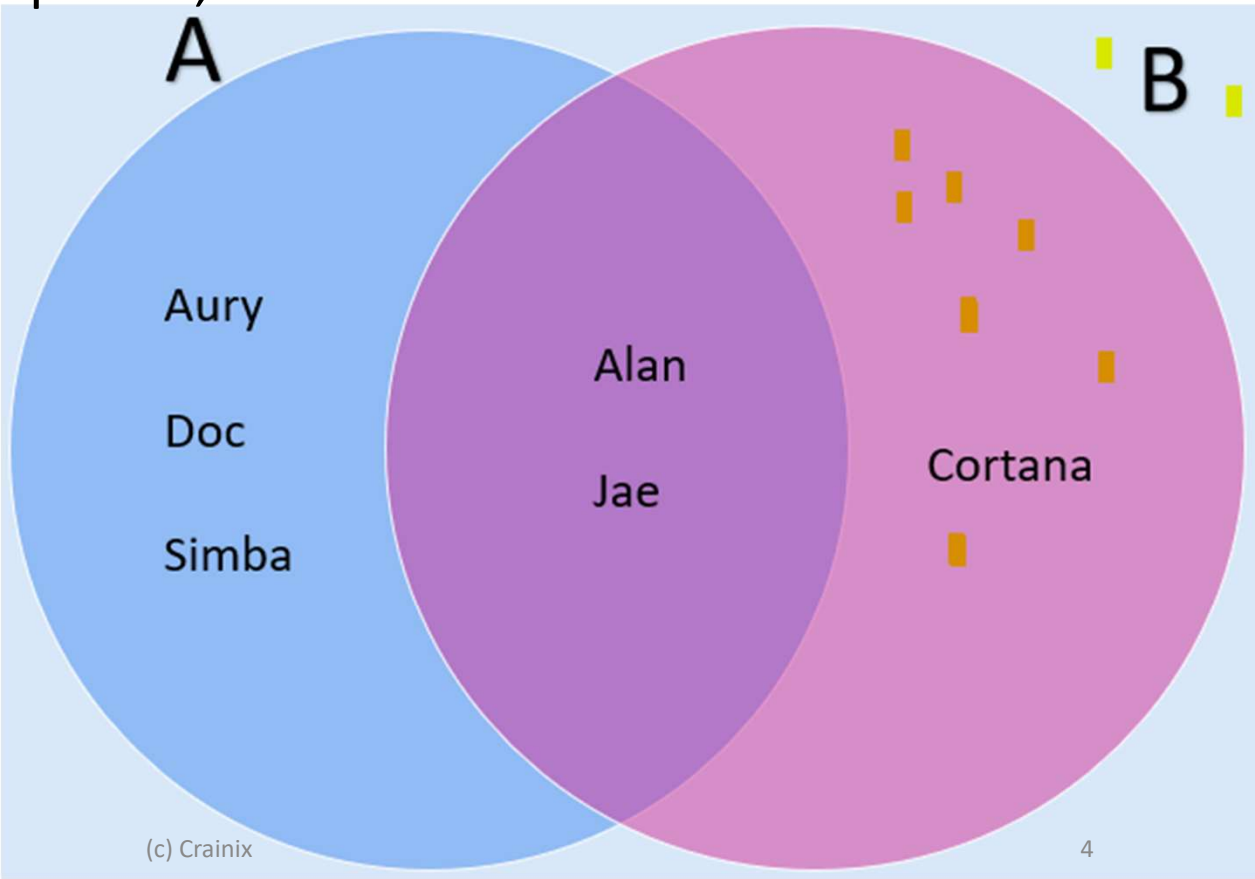
Set notation: read as the set of all x such that x is a member of team Geller. The colon means "such that" and ϵ is the symbol for member or element. {x: x ϵ Team Geller} {x: x ϵ

Set A is the mammals in our household.

Set B is the English speakers in our house.

Cortana was our Microsoft smart speaker,

like Siri or Alexa.



Set A is the mammals in our household.

Set B is the English speakers in our house.

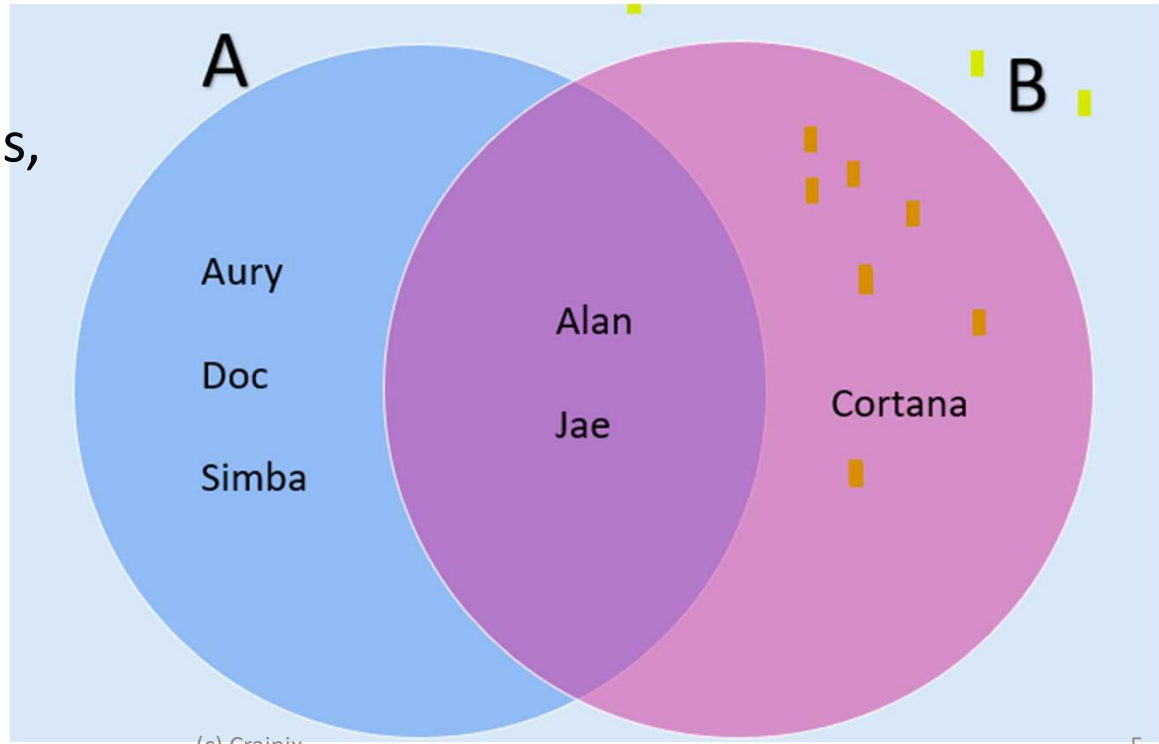
The **Union** of A and B or A union B (The union symbol is like a U.)

$A \cup B$={Aury, Doc, Simba, Alan, Jae, Cortana}

All the members of both sets, only written once.

Notice that the order doesn't matter.

$A \cup B = B \cup A$



Set A is the mammals in our household.

Set B is the English speakers in our house.

Intersection of A and B

or A intersect B

$A \cap B$={Alan, Jae}

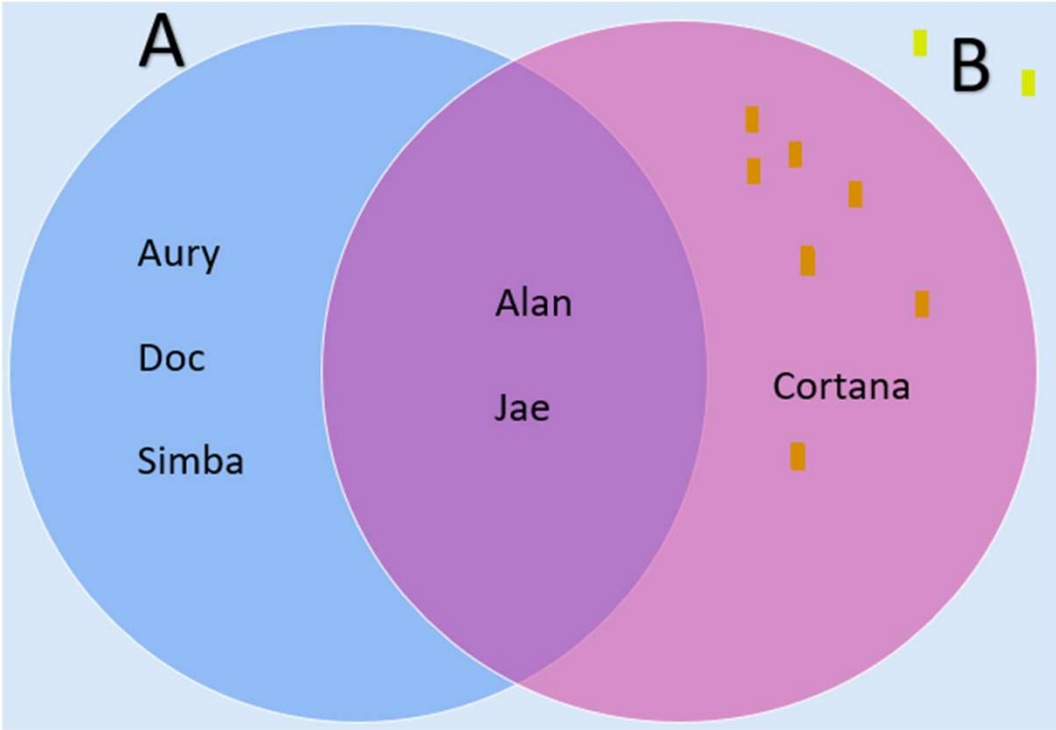
The elements that

both sets share.

The Overlap of the sets.

The order

doesn't matter



Set Theory | Introduction to College Mathematics (lumenlearning.com)

- The symbol $\in$ means "is an element of".
- A set that contains no elements, { }, is called the **empty set** and is notated $\emptyset$ or null set.
- A **subset** of a set A is another set that contains only elements from the set A.

If B is a subset of A, we write $B \subseteq A$

- A **proper subset** is a subset that is not identical to the original set—there is at least one element of the original set that is not in the subset.

If B is a proper subset of A, we write $B \subset A$

Set A is the set of all mammals in our household.
Set C is the set of animals in our household.
Set D is the set of all mammals in our town.

$A \subseteq C$

A is the same set as C

A is a proper subset of D

$A \subset D$

A is a subset of D but not the whole set.

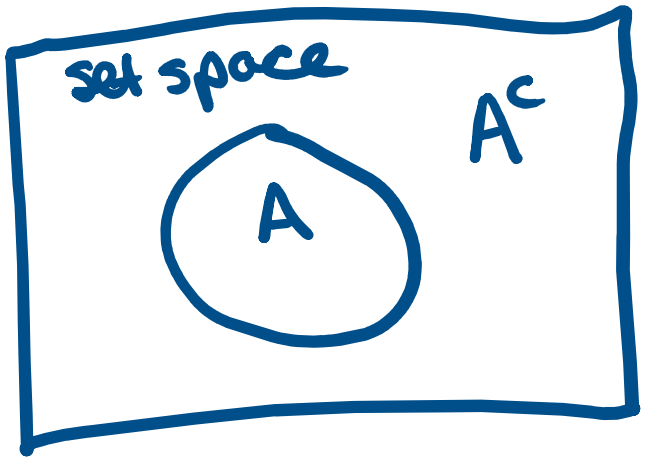
Union, Intersection, and Complement are the basic set operations

- The **union** of two sets contains all the elements contained in either set (or both sets). The union is notated $A \cup B$. More formally, $x \in A \cup B$ if $x \in A$ or $x \in B$ (or both)
- The **intersection** of two sets contains only the elements that are in both sets. The intersection is notated $A \cap B$. More formally, $x \in A \cap B$ if $x \in A$ and $x \in B$.
- The **complement** of a set A contains everything in the set space that is *not* in the set A. The complement is notated A', or A^c, *or sometimes* ~A.

Set Theory | Introduction to College Mathematics (lumenlearning.com)

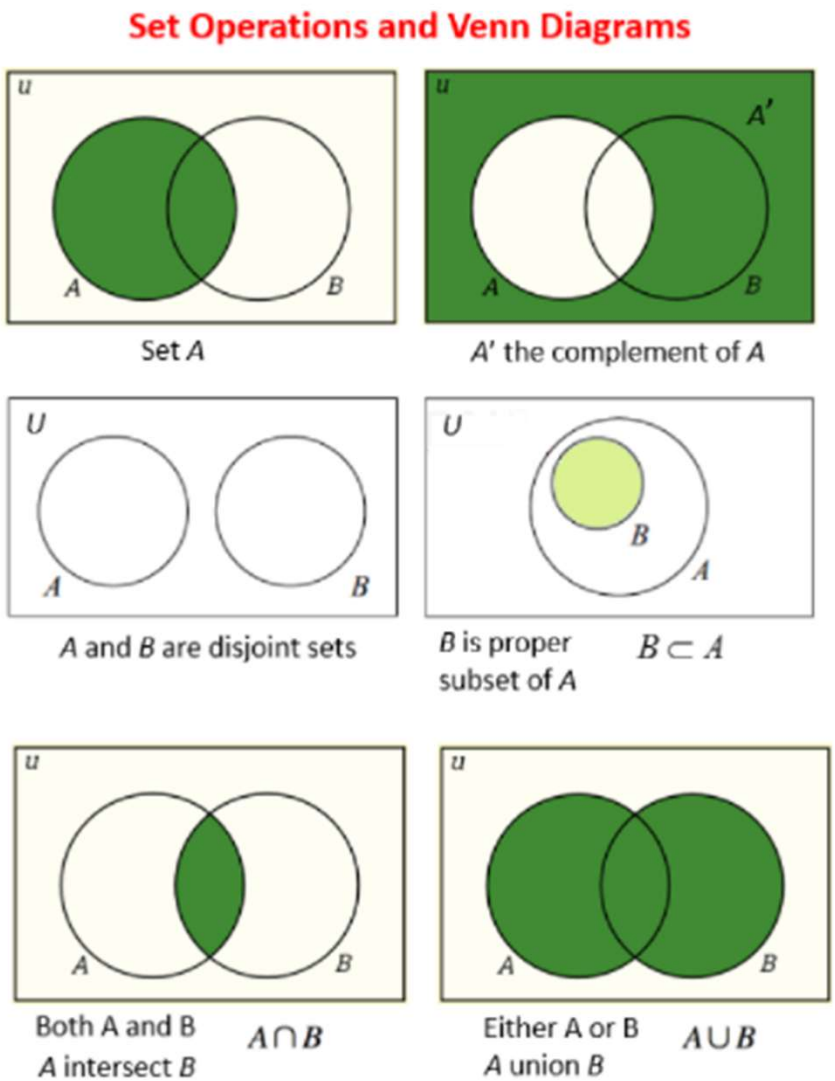
The **complement** of a set *A* contains everything in the set space that is *not* in the set *A*.

This is when we are talking about a set in a set space and not the universal set, which contains everything. This gets a bit in depth with proper terminology, so I am using what math textbooks usually use.

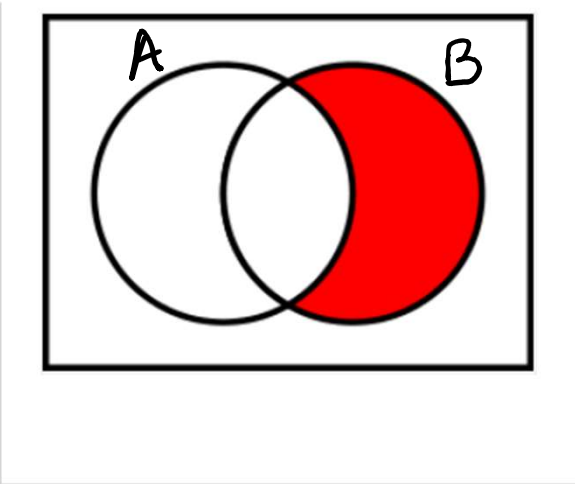


Set operations with Venn Diagrams

[Venn Diagrams And Subsets (video lessons, examples and solutions) (onlinemathlearning.com)]



Relative Complement or difference of sets is the x that is in one set but not another. Here it is the elements of B that are not in A.



Relative complement of A (left) in B (right) $A^c \cap B = B \setminus A$

Complement (set theory) - Wikipedia

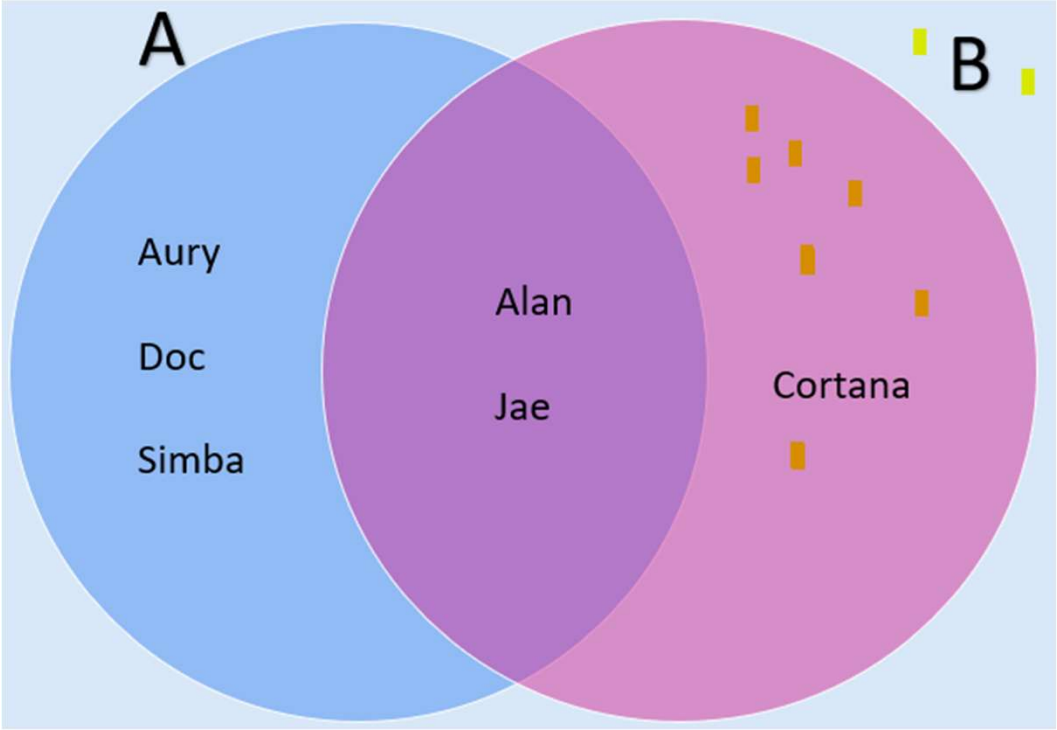
$$B \setminus A = \{x \in B \mid x \notin A\}$$

B-A is another way of writing the elements of B that are not in A. The elements of B minus the ones that are in A.

A-B is {Aury, Doc, Simba}

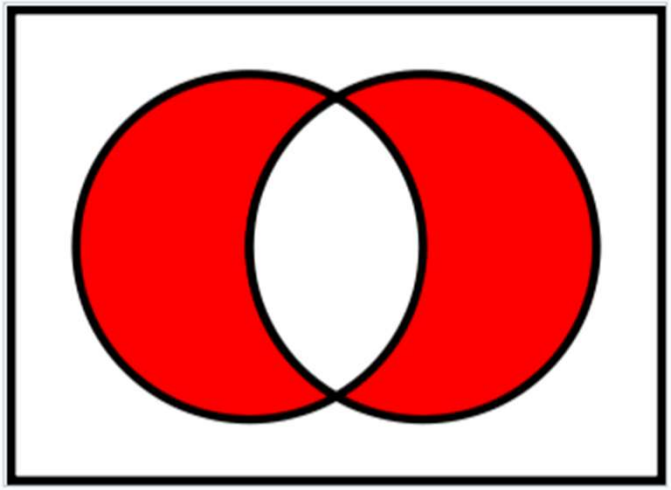
B-A is {Cortana}

Relative complement is
in the first and not in the second
Here the order matters as you
write it but not within the set.

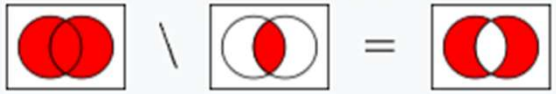


Symmetric difference of sets *A* and *B*, denoted $A \triangle B$ or $A \ominus B$,[7] is the set of all objects that are a member of exactly one of *A* and *B* (elements which are in one of the sets, but not in both).

$\{x : x \in A,\ x \in B,\ x \text{ not} \in A \cap B\}$



Venn diagram of $A \triangle B$. The symmetric difference is the union without the intersection:



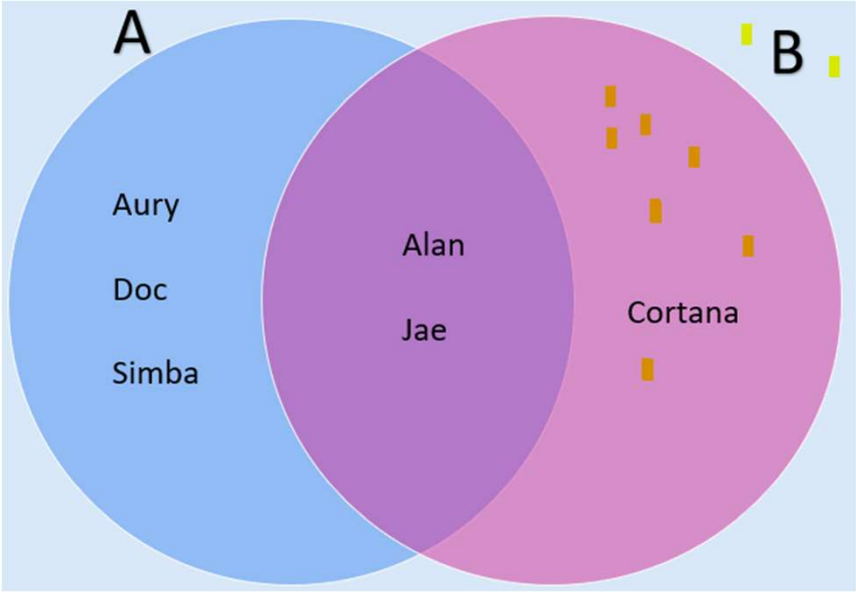
Symmetric difference - Wikipedia

Symmetric difference is in one or the other but not both.

AΔB={Aury, Doc, Simba, Cortana}

BΔA={Aury, Doc, Simba, Cortana}

With two sets, the order doesn't matter but with more than two sets, the order does matter. [Algebra 5 - Symmetric Difference – YouTube]



Set Operations for intersection and union are always simple but with more than two sets, some of the operations can get complicated.

Here is a video about operations with three sets. Venn Diagrams with 3 sets - Lesson - YouTube

We will come back to these concepts when we do probability and data science.